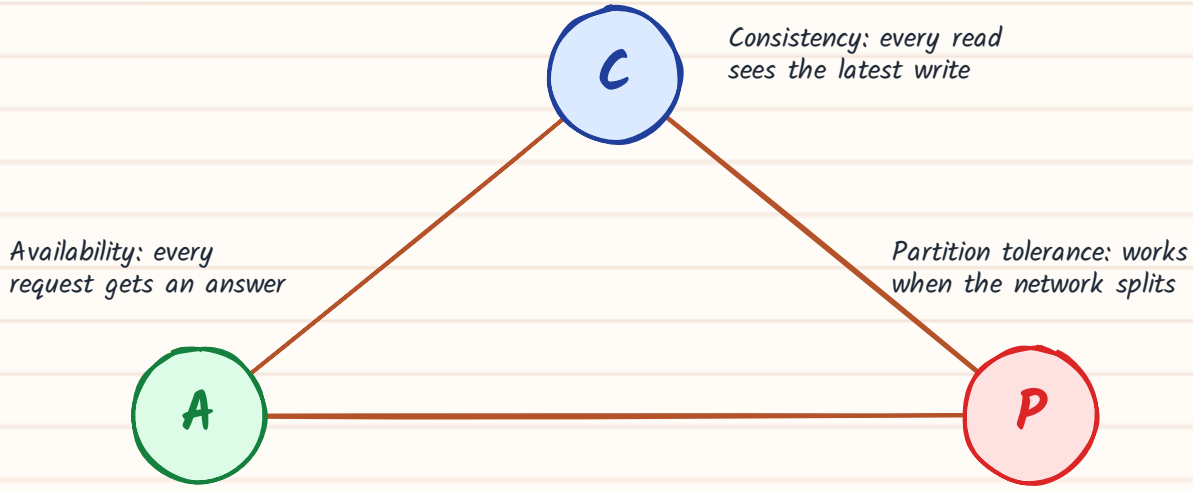
CAP, PACELC & Consistency

Lesson 5 · what CAP really says, and how to choose

In this lesson → ① Two ATMs ② CAP theorem ③ What CAP really says ④ CP vs AP ⑤ PACELC ⑥ Real databases ⑦ Consistency models ⑧ SnapLink choices ⑨ Interview Q&A
Running project: SnapLink, our photo-sharing app that grows from 1 server to 100M users.

1 The CAP theorem

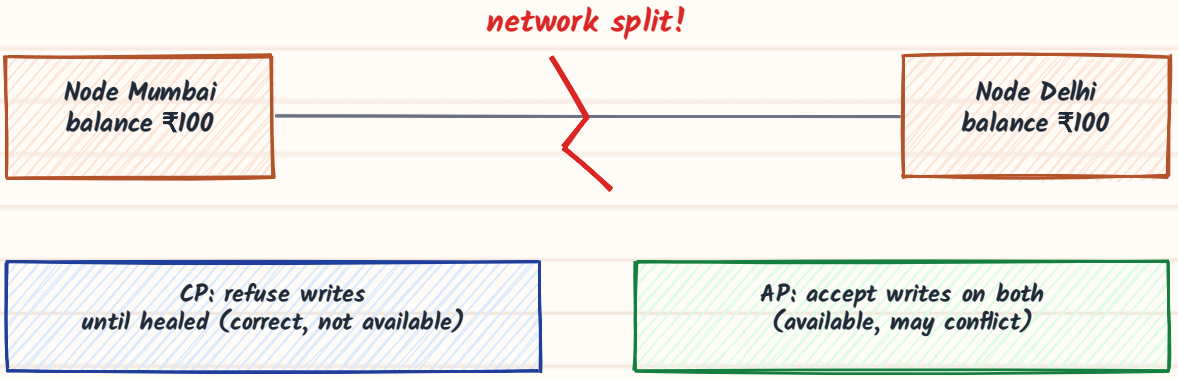


Letter	Means
Consistency	every read sees the most recent write (linearizable)
Availability	every request to a working node gets a non-error response
Partition tolerance	keeps working when the network between nodes breaks

Real meaning
Partitions are not optional in a distributed system. **During a partition**, choose C or A.

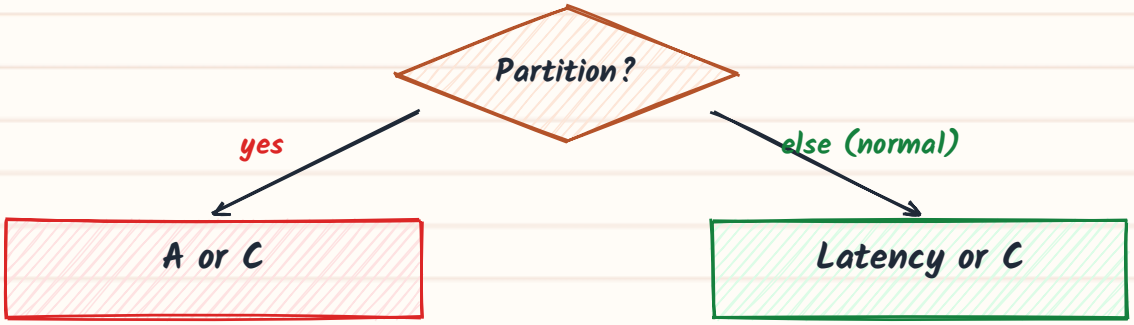
SYSTEM DESIGN NOTEBOOK

2 CP vs AP



	CP	AP
During partition	reject / block some requests	answer everything
Data	always correct	may diverge, reconcile later
Good for	payments, inventory, usernames	feeds, likes, carts, DNS

3 PACELC



most of the time the real trade-off is latency vs consistency

→ **P**artition → **A** or **C**

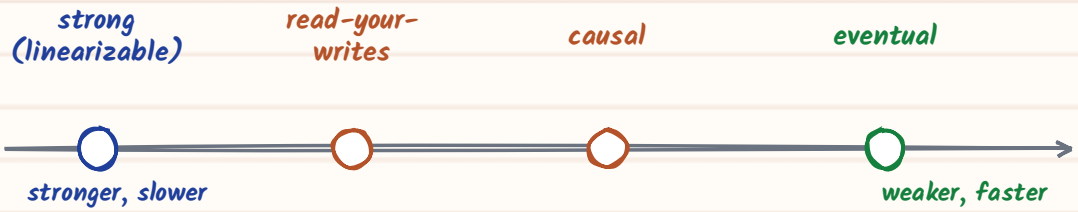
→ **E**lse → **L**atency or **C**onsistency

→ Sync replication = consistent but slower; async = fast but stale reads possible

4 Real databases

System	Leans to	Note
Postgres / MySQL (single leader)	C	failover → brief unavailability
Google Spanner	C	global, TrueTime clocks
Cassandra	A + low latency	tunable: ONE / QUORUM / ALL
DynamoDB	A + low latency	eventually consistent reads default; strong optional

5 Consistency models



Model	Guarantee	Example
Strong / linearizable	reads see the latest write, globally	bank balance
Read-your-writes	you always see your own writes	your new post
Monotonic reads	never see older data after newer	refresh doesn't go back
Causal	cause before effect for everyone	reply after its comment
Eventual	replicas converge if writes stop	like counts

6 Choosing for SnapLink

- ✓ Usernames, payments → strong
- ✓ Feed, likes, counts → eventual
- ✓ Own posts → read-your-writes
- ✓ Choose **per feature**

7 Interview Corner

Q. What does CAP say?

A. During a network partition, a distributed store must choose between consistency and availability.

Q. Why is "pick 2 of 3" misleading?

A. Partition tolerance is not optional; the C/A choice only applies during a partition.

Q. What does PACELC add?

A. Else (no partition), trade latency vs consistency.

Q. Strong vs eventual consistency?

A. Strong: every read sees the latest write. Eventual: replicas converge over time.

Q. Is Cassandra AP or CP?

A. AP by default, but consistency is tunable per query (e.g. QUORUM).

⚡ Revise in 60 seconds

- → Partition → **C or A**
- → Else → **L or C**
- → CP: banks, inventory
- → AP: feeds, carts
- → Strong → read-your-writes →
- → causal → eventual
- → Tunable (Cassandra, Dynamo)
- → Choose **per feature**

Next lesson → SD-06
How the web works: **DNS, TCP/UDP, HTTP, TLS.**
Watch on YouTube: **@TheCodeNotebook** · Notes: **thecodenotebook.com**